

# Annealed tiling for HSC + MBQ1 on the HSC-Niji wide footprint

D. Schlegel, 2026-10-08. Code: *skytiling (optimize\_tiling\_maps)*, data in *data/subaru/*.

**Setup.** The camera is described by per-sub-filter maps of relative throughput on the focal plane, built from Hironao Miyatake's MBQ1 dome flats (gain-corrected, sky-area-per-pixel and dome-illumination gradient divided out,  $\approx 1$  over the bulk of the field, 0 where the NO\_DATA mask flags the  $\sim 10.4'$ -wide filter-holder cross, the vignetted edge beyond  $46.5'$ , dead amplifiers and the two CCDs absent from the 413 flats). Every pointing is observed at four instrument rotations  $90^\circ$  apart, and the coverage of a sky point in sub-filter  $f$  is the sum of throughput over all exposures. Simulated annealing moves each pointing centre ( $\leq 0.75^\circ$ ) and the rotation offset of its 4-PA set to minimise the summed per-filter coverage variance over 4M random points inside the  $1194 \text{ deg}^2$  footprint ( $0.75^\circ$  discs around the 802 HSC-SSP Wide centres at  $-6^\circ < \text{Dec} < 6.4^\circ$ ). Two starting points with the same exposure budget as Atsushi's phase-1 pattern (4 dithers  $\times$  802 field centres  $\times$  4 PAs = 12,832 exposures): that pattern itself (3208 pointings, 300 iterations), and a Fibonacci lattice of 3207 pointings (277 iterations). Each run took  $\sim 6$  s per iteration on one Perlmutter node with 64 workers (step size  $0.1^\circ$ ,  $10^\circ$  shrinking by 0.99 per iteration).

**Result** (footprint interior, 68% of the area, away from the stripe edges; coverage = effective number of full-throughput exposures):

|                                            | 413                                      | 439                                      | 465                                      | 490                                      |
|--------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|
| Atsushi phase 1: mean $\pm$ rms (rms/mean) | $2.85 \pm 1.19$ (0.42)                   | $3.24 \pm 1.12$ (0.34)                   | $3.24 \pm 1.16$ (0.36)                   | $3.11 \pm 1.10$ (0.35)                   |
| fraction $< 1.5 / < 0.5$                   | 13.0% / 2.1%                             | 6.5% / 0.8%                              | 7.4% / 1.2%                              | 7.7% / 0.9%                              |
| Fibonacci lattice, un-annealed             | $2.79 \pm 1.08$ (0.39)                   | $3.19 \pm 1.08$ (0.34)                   | $3.17 \pm 1.15$ (0.36)                   | $3.06 \pm 1.04$ (0.34)                   |
| <b>Annealed from Atsushi phase 1</b>       | <b><math>2.60 \pm 0.87</math> (0.34)</b> | <b><math>2.98 \pm 0.87</math> (0.29)</b> | <b><math>2.95 \pm 0.85</math> (0.29)</b> | <b><math>2.86 \pm 0.84</math> (0.29)</b> |
| fraction $< 1.5 / < 0.5$                   | 9.7% / 0.7%                              | 4.4% / 0.2%                              | 4.3% / 0.2%                              | 5.3% / 0.3%                              |
| <b>Annealed from Fibonacci lattice</b>     | <b><math>2.56 \pm 0.86</math> (0.34)</b> | <b><math>2.94 \pm 0.88</math> (0.30)</b> | <b><math>2.91 \pm 0.85</math> (0.29)</b> | <b><math>2.81 \pm 0.84</math> (0.30)</b> |
| fraction $< 1.5 / < 0.5$                   | 10.2% / 0.8%                             | 4.9% / 0.3%                              | 4.7% / 0.2%                              | 5.8% / 0.3%                              |

For the same number of exposures the annealing lowers the rms of the coverage by 24–28% in every filter (rms/mean by 14–20%), cuts the deep holes (coverage  $< 0.5$ ) by a factor 3–6, and removes the periodic pattern of Atsushi's layout (Figure 1). The mean is 8–10% lower because pointings near the stripe boundary move outward to flatten the edge and the coverage they put outside the footprint is not counted; in the interior the annealed solutions are more uniform at every quantile. The two starting points converge to the same solution quality — the convergence curves lie on top of each other (Figure 2) and the final statistics agree to within  $\sim 1\%$  — with the Atsushi start marginally ahead (1.5% higher mean, slightly fewer holes), so the result is set by the camera masks and the tile density, not by the starting layout. 413 remains the worst filter in every case because of its two missing CCDs. The 0/1 (G00D) and throughput-weighted statistics differ by only 2–4%.

**Rotations.** Allowing the 4-PA set of each pointing to turn is worth a few per cent in rms on top of the position moves. The annealed offsets (Figure 2, lower left) have an rms of  $17\text{--}19^\circ$  in both runs, with 57–59% of pointings turned by more than  $10^\circ$  and 9–10% by more than  $30^\circ$ ; the distribution is broad and single-peaked with a small positive mean ( $+3^\circ$ ), i.e. the solution does not favour a particular grid of angles. The pointings moved by a median  $0.19\text{--}0.20^\circ$  from their starting positions (max  $0.75^\circ$  = the cap, reached by a handful), so in both cases the final layout is a modest perturbation of the start.

**Convergence.** The objective (sum over filters of rms/mean) fell from 1.58 to 1.24 (Fibonacci start) and from 1.64 to 1.24 (Atsushi start); 85% of the gain came in the first 40 iterations, and the last 100 iterations gained  $< 0.01$  (Figure 2, upper left). At the final slope of  $7\text{e-}5$  per iteration, another 300 iterations would buy well under 1%, so both runs are converged for practical purposes at this tile density. Larger gains would come from more tiles (lower rms/mean at higher mean coverage) or from a different objective (e.g. penalising holes rather than variance).

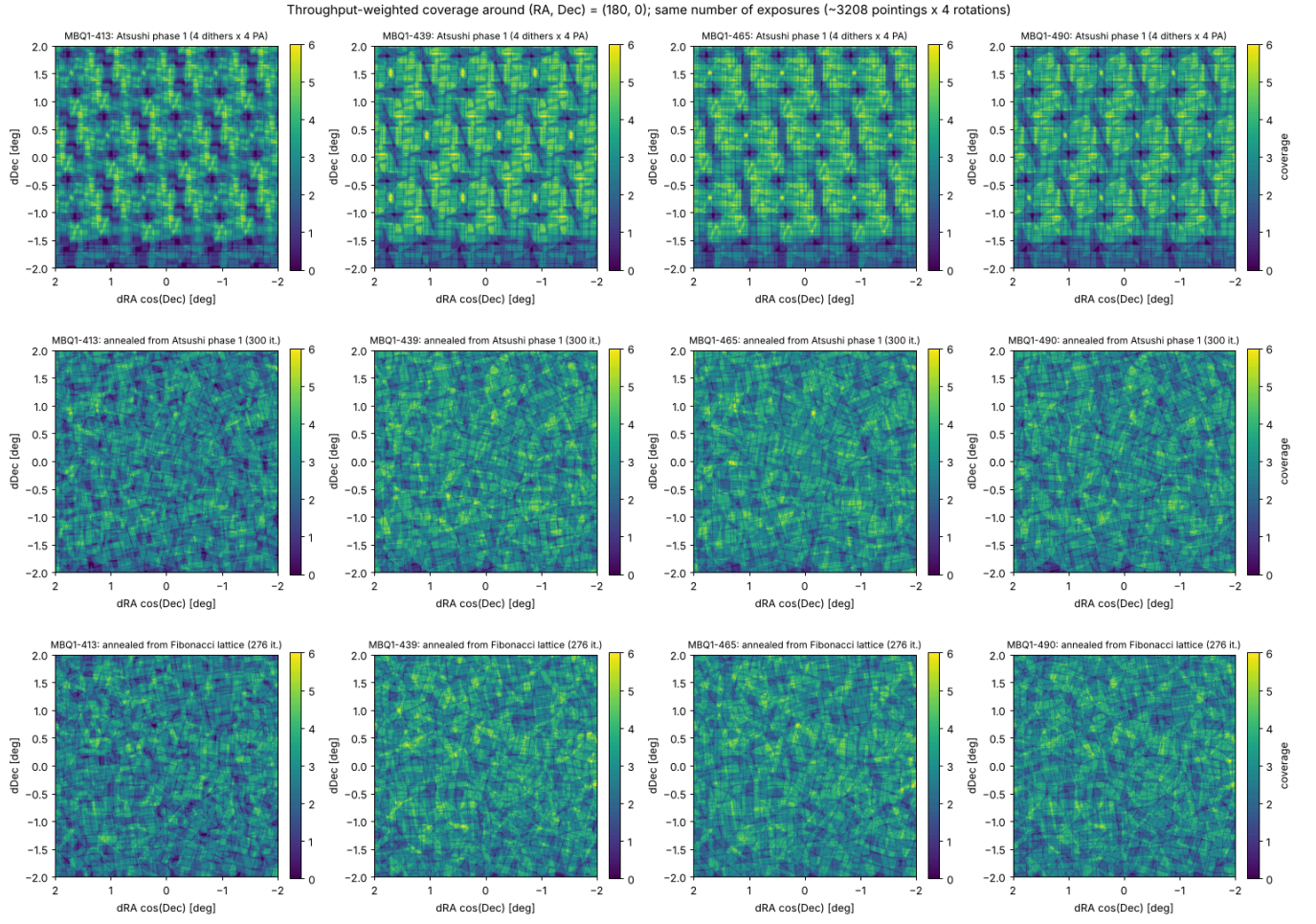


Figure 1. Throughput-weighted coverage per sub-filter in a  $4^\circ \times 4^\circ$  field around (RA, Dec) = (180°, 0°) for Atsushi's phase-1 pattern (top), the tiling annealed from it (middle, iteration 300) and the tiling annealed from the Fibonacci lattice (bottom, iteration 276), same colour scale.

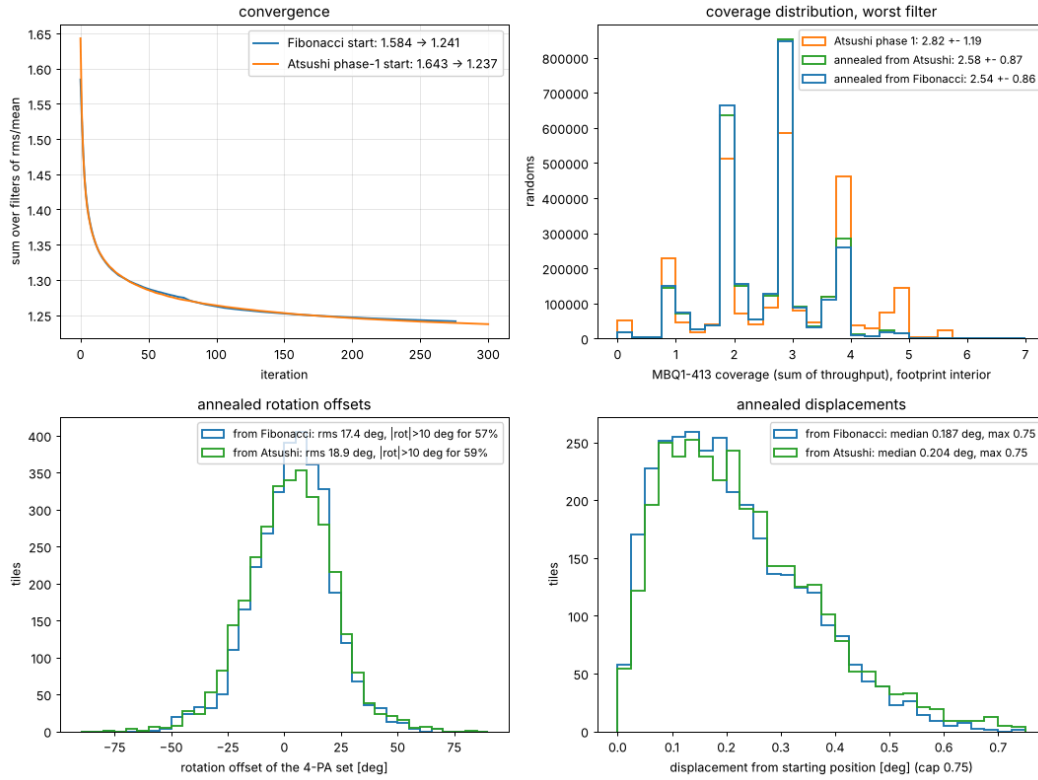


Figure 2. Convergence of the objective for both starts; distribution of MBQ1-413 coverage over the footprint interior; annealed rotation offsets; and displacement of each pointing from its starting position.

**Robustness and precision.** The optimizer fits to one realisation of 4M randoms, so its statistics on that set are slightly optimistic: on an independent set (4M or 8M, which agree with each other to the third digit) the final Atsushi-start solution has interior rms/mean 0.348 / 0.305 / 0.298 / 0.307 and 10.4% / 5.0% / 4.9% / 6.0% below 1.5, i.e. ~3% worse than on the training randoms, while Atsushi's original pattern (not fitted) is unchanged at 0.42 / 0.35 / 0.36 / 0.36. The honest improvement is therefore 25% in rms (17–20% in rms/mean) and a factor 2.3–4 in deep holes. Rounding the solution changes nothing measurable: RA, Dec to 3 decimals (1.8") and ROT to 1 decimal (0.1°) shift the summed rms/mean by  $2 \times 10^{-5}$ ; even 2 decimals / 0.5° rounding costs only  $1.6 \times 10^{-3}$  (0.1%). The delivered table uses 3 decimals in RA/Dec and 1 in ROT.

Files: data/subaru/anneal/nersc\_atsushi/hsc\_mbq1\_tiling\_atsushi\_final.{fits,csv} (the rounded final solution: tileid, ra, dec, rot), data/subaru/anneal/nersc\_atsushi/tiles\_{0000,0300}.fits and data/subaru/anneal/nersc\_fibonacci/tiles\_{0000,0276}.fits (initial and final; columns ra, dec, rot, ra0, dec0; each row is one pointing to be observed at rot, rot+90, rot+180, rot+270). The mapping of rot onto INSR0T\_PA (sign and zero point) still has to be calibrated against the instrument.